

Publisher Submission Sheet

Book Cover & Manuscript Metadata

Book Cover

Main Author Name

Kundai Farai Sachikonye

Co-Authors' Names

max. 2

None — single-authored monograph.

Book Title

100-character limit

The Blind Leading the Blind

[27 characters / 100 limit]

Subtitle

100-character limit

Categorical Resolution of Biological Dynamics: Cellular State, Disease, and Therapy

[83 characters / 100 limit]

Spine

60-character limit

The Blind Leading the Blind Sachikonye

[40 characters / 60 limit]

Blurb

2200-character limit; min. 250 words

In 1568 Pieter Bruegel painted six blind men falling into a ditch, each with a different, clinically diagnosable eye disease. This monograph takes that image as a theorem about biology: every finite observer—every instrument, every receptor, every cell—is blind in its own specific way, and no ensemble of blind observers corrects itself by aggregation. Yet a healthy body still navigates. How?

From just two axioms—that physical systems occupy a bounded phase space, and that any finite observer partitions it into distinguishable cells—the book derives a complete, self-contained mathematical framework for cellular state, disease, and therapy. It proves that every bounded receiver has a strictly positive epistemic “floor”: the truth about a state is never a point but a region of positive measure. From this follow the No Template Theorem (no cell can store a model of itself, so self-diagnosis is structurally impossible), the Group Blindness Theorem (collective observation cannot reach zero error), and the Local Invisibility Theorem (disease is invisible to any observer confined to a single reaction).

The resolution is geometric. Health is not knowledge of health but *self-consistency*: when the reaction rates around every metabolic loop close on themselves—zero holonomy—the circuit is well, and this can be checked from the inside with no external template. Therapy becomes the restoration of loop closure, computed as a sparse, polynomial-time optimisation. Purpose itself appears as a dynamical fixed point: the destination is enacted, not seen.

Rigorous yet vivid, moving from Renaissance painting to category theory, differential topology, and convex optimisation, this is a first-principles account of why living systems work despite being built entirely from blind parts.

[1799 characters / 2200 limit] [266 words]

Author's Biography

500-character limit; min. 70 words

Kundai Farai Sachikonye is a researcher in bioinformatics at the Technical University of Munich and the AIME Registry for Artificial Intelligence. He develops first-principles mathematical frameworks that unify observation, computation, and physical dynamics, focused on the categorical geometry of cellular state, disease, and therapy. His research asks how living systems stay well while built entirely from bounded, error-prone parts. This present book is the synthesis of that programme of work.

[499 characters / 500 limit] [70 words]

Manuscript

Book Language

English (British)

Book Category

Science » Life Sciences » Biophysics & Mathematical Biology

Secondary: Mathematics » Applied Mathematics (Dynamical Systems, Optimisation); Medicine » Systems Biology & Theoretical Medicine

Keywords

- categorical dynamics; partition geometry; bounded observers
- epistemic floor; cell-truth; resolution limit
- metabolic circuits; Wegscheider holonomy; loop closure
- disease as coherence deficit; No Template Theorem
- self-consistency; reference-free diagnosis
- sparse ℓ_1 therapeutic optimisation; drug combination
- oscillator coherence; Kuramoto synchronisation
- mathematical biology; theoretical medicine; systems biology